Course Title: Mechanics and Philosophy of Large Language Models
Course Number: CS 4973, CRN: 18451
Semester: Fall 2025
Credits: 4

Instructors:
Meica Magnani ("Me-kuh"), she/her (m.magnani@northeastern.edu)
Felix ("Fee-licks") Muzny, the/he (f.muzny@northeastern.edu)
you can call both instructors by their first names

This course can be used to fulfill the same course requirement that Tech and Human Values fulfills.

Course Description:
Covers the internal components of large language models (LLMs), the data and technological systems necessary to build and deploy them, and the real-world applications, effects, and lines of philosophical inquiry around these models. Key concepts include computational knowledge representation, data and scale, computing infrastructure, opportunities and limitations posed by such models, and key threads of ethical inquiry surrounding them. Students will learn the history of LLMs as situated in the history of computing and automatic translation technologies. Students will study the practical capabilities and limitations of these models, explore the various contexts in which they are currently deployed, and explore the technological, societal, and environmental impacts of these technologies. This course does not teach students to program, nor does it expect students to program.

Learning Outcomes:
By the end of this course, students will:
- have a technical understanding of how language models work and what their current limitations are
- have a critical understanding of the role that data and scale play with these models
- be able to deploy philosophical and ethical tools, concepts, and methodologies to analyze and understand the ethical and social dimensions of LLMs.
- be able to deploy philosophical and ethical tools, concepts, and methodologies to develop, articulate, and defend one's own stance on social and ethical issues related to LLMs.

Classroom Expectations:
Students are expected to come to class ready to participate–this course is heavily centered on small- and large-group work, discussion, and critically engaging with assigned readings.

AI Policy:
This is a course that specifically talks about LLMs and AI! It would be hypocritical for us to have a blanket "no AI" policy. There will be times we explicitly ask you to use LLMs for part of your work. In all other cases, the policy that we will write together in class applies.

Written in class 9/5/2025

How are LLMs allowed to be used in this course?
- To help generate ideas, get pointers for more information, start research, experimentation

How do you indicate to instructors where you used LLMs?
- Include footnotes with prompts for anything that gets copy+pasted, including which model it came from
- Indicate if any sources came from an LLM with footnotes

- Indicate if an idea came from an LLM with a footnote
- When responding to readings, note in a comment if you used an LLM

How are LLMs **not** allowed to be used in this course?

- To fully generate code
- Generate answers to questions (written or otherwise)
    - re-engaging with readings and formulating answers in your own words have strong benefits to learning, make sure to do this *before* turning to LLMs
- Generate essays
- Discussing ethics
- Summarizing reading **without also** doing the reading

On a scale from 1 (least permissive) to 5 (most permissive), this course is a **3.5**, meaning that LLM use in specific context and for specific tasks is generally okay. LLMs should be used as *aides* rather than crutches. LLMs should not be used to generate full answers. All topics need to be understood and analyzed by students even if an LLM was also used.

If you use an LLM in a way that does not follow this policy, you will be required to re-do the assignment with no LLMs and extra components as noted by the instructors. If this happens more than once, you will receive a 0 on the assignment and the instructors will submit a report to OSCCR.

Collaboration policy:
All work that you submit should be your own. In the case of group work, it should be the work of you and the classmates you worked with. Feel free to discuss the topics we're addressing in class with your friends, but don't have them do your work for you. If you turn in work that is not your own, you will receive a 0 on the assignment in question.

Course Work:
The work that you do for this course will be composed of the following elements:

1) Readings – primarily completed ***outside*** of class. Readings associated with a particular day need to be completed ***before*** class begins. When readings are not completed, this will impact your participation grade.
2) Participation – your participation grade will be based on your level of engagement with the material in class. It is not simply an attendance grade. See below for our policy when you are sick.
3) Portfolios – primarily completed ***in*** class. You will add to your portfolio throughout the semester and will be turned in three times: September 30th, October 31st, and December 5th.
4) Final Projects – primarily completed ***outside*** of class.

Late Policy:
All work must be turned in on time unless you have requested an extension ***ahead of time if at all possible***. To request an extension, email Felix (f.muzny@northeastern.edu) with a brief explanation and the specific time you are requesting the extension until. Extensions are primarily granted based on ***proactive communication***. If you turn in an item late and did not request an extension, you will receive a 10% penalty on the assignment in question.

Sick Policy:
If you are sick, do not come to class. You are always welcome to wear a mask in class regardless of your health status. If you think you may be getting sick but are well enough to attend class, we ask that you wear a mask, similarly if you are living with someone who is sick.

To make up the work that you miss on the day that you are sick, email Felix (f.muzny@northeastern.edu) as soon as you can to arrange make-up work. We do not require doctor's notes.

Schedule (Subject to change at instructors' discretion):
(Up to date version: cs4973_calendar_fall2025)

The up-to-date version of this calendar is linked above and from the course website; find an *initial draft that is subject to change* below.

Week	Weekday	Date	Topic	Readings All readings to be completed *before* corresponding lecture
1	Friday	9/5/2025	What is AI? What is philosophy? How can it help us think about LLMs? the role of AI and LLM in human life?	
2	Tuesday	9/9/2025	History of Natural Language Processing, how computers represent language, how do LLMs work (part 1)?	https://en.wikipedia.org/wiki/ASCII
	Friday	9/12/2025	What is it to understand a language?	Reading: Searle's Chinese Room
3	Tuesday	9/16/2025	How do LLMs work (part 2)? What are "non-determinism" and "reinforcement"?	Readings for remaining weeks will be on the up-to-date schedule as the semester progresses
	Friday	9/19/2025	Guest speaker: Shane Steinert-Threlkeld	
4	Tuesday	9/23/2025	Data data data, scale, privacy, and more data	
	Friday	9/26/2025	Conceptions of the Mind: where does AI fall?	
5	Tuesday	9/30/2025	Why is math so hard for LLMs?	
	Friday	10/3/2025	Common Ground, Reasoning in Conversation, Speech Act Theory: What is it to use a language? What is implicature?	

6	Tuesday	10/7/2025	Lying, Truth, and Bullshit	
	Friday	10/10/2025	Why do LLMs hallucinate? What safe-guards do we have?	
7	Tuesday	10/14/2025	TBD: Creativity and mental models	
	Friday	10/17/2025	TBD: Creativity and mental models	
8	Tuesday	10/21/2025	AI Companion Bots: what's going on?	
	Friday	10/24/2025	Can an LLM have a "personality"? Why?	
9	Tuesday	10/28/2025	Discriminatory Harms and LLMs: Allocative, Representational, and Aspirational	
	Friday	10/31/2025	How does the internet work and why do we care?	
10	Tuesday	11/4/2025	Ethics of climate change and distributive justice	
	Friday	11/7/2025	LLMs and the environment	
11	Tuesday	11/11/2025	NO CLASS VETERANS DAY	
	Friday	11/14/2025	More on LLMs + Safeguards	
12	Tuesday	11/18/2025	AI: Creativity and Future of Work	
	Friday	11/21/2025	TBD	
13	Tuesday	11/25/2025	Future of Education	
	Friday	11/28/2025	NO CLASS THANKSGIVING BREAK	
14	Tuesday	12/2/2025	TBD	
	Friday	12/5/2025	Presentations (attendance mandatory)	
15			This is finals week. You will have some work to do!	

Grading:

Students will be assessed on the following components:

Component	Percent
Portfolios	40%
Participation	20%
Final Project	40%

Final grades will be based on the following scale. Decimals will be rounded to the nearest whole number.

Letter	Range (percent)
A	95 - 100
A-	90 - 94
B+	87 - 89
B	83 - 86
B-	80 - 82
C+	77 - 79
C	73 - 77
C-	70 - 72
D	60 - 69
F	< 60

Accommodations:

It is our job to create a classroom environment that is most conducive to you learning well. If you have accommodations from the Disability Resource Center, please provide your letter to us early in the semester so that we can arrange for these accommodations. If you wish to receive accommodations and do not have a letter, please visit the DRC at 20 Dodge Hall or call (617) 373-2675.

Student Names and Pronouns:

- If your name and pronouns are not in alignment with those listed on our class roster, please let us know either in person or via email so that we can ensure you are correctly addressed in this class.
    - If you wish to add, change, or update your pronouns in Canvas (for courses that do have Canvas websites), go to "Account" > "Profile" > "Edit Profile", then add, change, or update your pronouns and display name.
    - If you wish to change or update your name here at Northeastern as a whole, find instructions with the registrar here.

- We recognize that your legal information doesn't always align with how you identify. Students may update their first and middle names as well as gender marker with the registrar, even if they are not your legal names or gender marker. Those names and gender marker are what would appear publicly in most university systems. In the absence of such updates, what we see on most university systems by default are your legal name and gender marker.

Classroom Environment:

To create and preserve a classroom atmosphere that optimizes teaching and learning, all participants share a responsibility in creating a civil and non-disruptive forum for the discussion of ideas. Students are expected to conduct themselves at all times in a manner that does not disrupt teaching or learning. Your comments to others should be constructive and free from harassing statements. You are encouraged to disagree with other students and the instructor, but such disagreements need to be respectful and be based upon facts and documentation (rather than prejudices and personalities). The instructor reserves the right to interrupt conversations that deviate from these expectations. Repeated unprofessional or disrespectful conduct may result in a lower grade or more severe consequences.

Part of the learning process in this course is respectful engagement of ideas with others.

The Code of Student Conduct can be found on the OSCCR website.

Title IX:

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.

Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking.

The Title IX Policy applies to the entire community, including male, female, non-binary, and transgender students, faculty and staff.

If you or someone you know has been a survivor of a Prohibited Offense, confidential support and guidance can be found through University Health and Counseling Services staff and the Center for Spiritual Dialogue and Service clergy members.

By law, those employees are not required to report allegations of sex or gender-based discrimination to the University.

Reports can be made non-confidentially to the Title IX Coordinator within the Office for Gender Equity and Compliance at: titleix@northeastern.edu and/or through NUPD (Emergency 617.373.3333; Non-Emergency 617.373.2121).

Reporting Prohibited Offenses to NUPD does NOT commit the victim/affected party to future legal action.

Faculty members are considered "responsible employees" at Northeastern University, meaning they are required to report all allegations of sex or gender-based discrimination to the Title IX Coordinator.

In case of an emergency, please call 911.

Please visit http://www.northeastern.edu/titleix for a complete list of reporting options and resources both on- and off-campus.

Religious Holidays:

The course staff will make every effort to deal reasonably and fairly with all students who, because of religious obligations, have conflicts with scheduled exams, assignments or required attendance. In this class, contact the course staff at least 7 days in advance of the conflicting date to reschedule a homework or quiz due date.